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Design and CFD-Based Optimization of Multi-Slit Pintle Injector for High Pressure LOx-Methane Rocket Engine

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Abstract

Increase in demand for sustainable technologies has accelerated the use of environmentally friendly propellants in rocket propulsion. Liquid oxygen (LOx) and methane (CH_4) have gained attention as a green alternative to hypergolic propellants due to their high performance, non-toxic nature, and in-situ resource utilization potential. Injector plays a key role in governing mixing, combustion efficiency, and stability. Throttling capability is essential for reusable launch vehicles and planetary landers, where precise thrust control is required. Pintle injectors address these needs by offering smooth throttling, reduced combustion instabilities, and reliable operation. This study presents design and analysis of an oxidizer centred pintle injector for a high-pressure LOx-methane engine, with the objective of optimizing the pintle design through CFD simulations. An analytical procedure to design multi-slit pintle injector is presented and combustion modelling at supercritical conditions is carried out to study the influence of key geometric parameters. Flamelet Generated Manifold (FGM) combustion model with the Soave Redlich Kwong (SRK) real fluid EoS is used in a RANS framework. Results show that injector geometry strongly influences mixing behaviour. Variations in annular gap and skip ratio have no substantial impact on mixing performance, whereas pintle geometric parameters play a dominant role in mixing. Decrease in the number of slits leads to improved mixing efficiency. Further, the single row slit configuration enhances mixing even at higher slit numbers as the interaction area between the LOx and methane significantly increases. This work offers systematic insight into pintle injector design optimization for LOx-methane propulsion systems and supports the development of efficient and sustainable rocket engines.

Keywords: LOx-methane, Pintle Injector, Supercritical Combustion, FGM.

1 Introduction

The growing global emphasis on sustainability and environmentally responsible technologies has significantly influenced the direction of modern space exploration. Conventional propellants such as hydrazine based hypergolic propellants and RP-1 (kerosene) have served reliably for decades; however, their toxic nature, handling hazards, and limitations in reusability have motivated the search for cleaner and more sustainable alternatives. In this context, liquid oxygen (LOx) and methane (CH_4) have emerged as a highly promising propellant combination for next-generation rocket engines[1]. LOx-methane systems offer several key advantages: higher specific impulse compared to kerosene, non-toxic and environmentally friendly operation, reduced coking tendencies in cooling channels, and simplified ground handling logistics[2]. Moreover, methane's ability to be produced in-situ on extra-terrestrial bodies, particularly Mars, through the Sabatier process [3], enhances its attractiveness for future long duration missions and planetary exploration. These attributes position LOx-CH_4 propulsion systems as central to the development of sustainable and reusable spaceflight technologies [4].

The efficiency of a liquid rocket engine is highly dependent on the performance of its injector, which governs propellant atomization, mixing, and combustion stability [5]. The injector ensures uniform propellant mixing, which boosts combustion efficiency and prevents localized hot spots, thereby protecting the thrust chamber from damage. Among the available injector configurations such as showerhead, coaxial swirl, and impinging jet designs the pintle injector has attracted particular attention due to its unique advantages like precisely modulating flow path geometry and pressure drop [6]. The pintle injector offers stable deep throttling capability without significant loss of efficiency, reduced susceptibility to both low and high frequency combustion instabilities, reliable performance [7]. Pintle injector is structurally simple design compared to multi-element injectors that lowers cost and manufacturing complexity, and a proven heritage demonstrated in the Apollo program [8] with renewed application in modern reusable launch vehicles [9]. These attributes make them particularly suitable for modern LOx-methane propulsion systems, especially where reusability, throttling, and reliability are critical [10].

The pintle injector, first explored at JPL in the mid-1950s and formally developed by TRW in 1960, became publicly available with its 1972 patent release [11]. Its breakthrough application in the Apollo LMDE enabled 10:1 thrust variation with stable combustion and precise control, establishing the pintle injector as the benchmark for throttle-able liquid rocket engines. A pintle injector is a throttle-able injector configuration in which a central pintle element regulates the flow distribution between propellants. In the oxidizer centered type, the oxidizer is injected through the central passage, while the fuel is introduced through an annular gap surrounding the pintle. As the high momentum central oxidizer jet interacts with the annular fuel stream, strong shear and turbulence are generated at the interface, resulting in efficient rapid mixing [12]. This mechanism ensures enhanced combustion performance and stability. Pintle injectors can be designed in three ways: the continuous type [13], multi hole [14] and multi slit type [15].

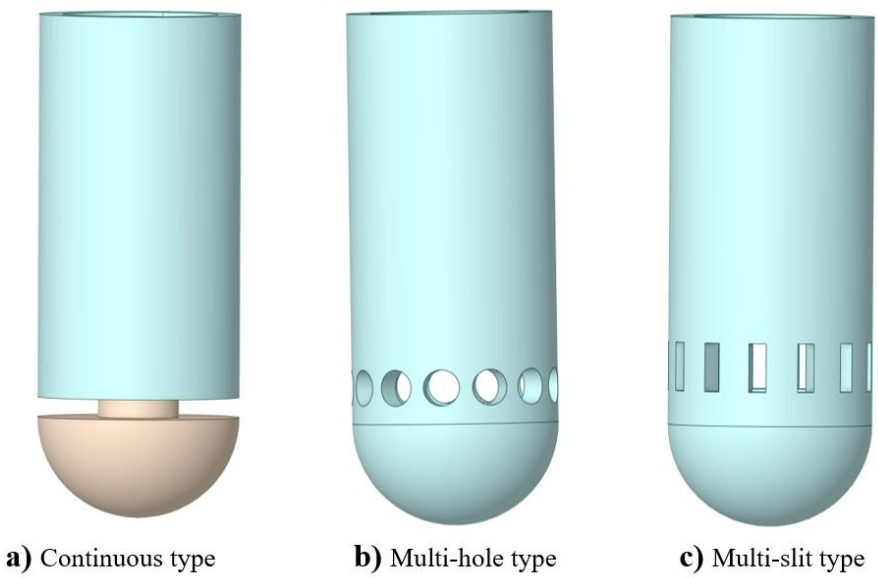

Fig.1 Different types of pintle configurations

Fig.*1*shows the three different configurations of pintle injector. Recently, Son et al.[16]studied the combustion characteristics of a continuous type pintle injector using a 6-step JL mechanism with the EDC model and found that incomplete combustion at low mass flow rates could be mitigated by reducing the pintle opening distance, reducing the injection area. Fang et al.[17], conducted numerical analysis for a continuous type pintle injector, study mentions that four skip ratios were chosen: 0.25, 0.5, 1.0, and 1.5 among this the skip ratio 1.0 gives the highest efficiency ~95.8%. The study highlighted the importance of skip distance(L_s) generally referred as the axial distance that the fuel travels from the exit of annular gap along the pintle post before it gets in contact with the oxidizer stream. The study states that optimum skip distance enhances better mixing performance. Hwang et al.[18]experimentally investigated two row multi slit pintle injector configurations under supercritical operating conditions. Nine configurations were tested with blockage factors (BF),defined as the ratio of the annular flow area blocked by the central pintle oxidizer post to the total annular flow area, ranging from 58 ± 7% and Total Momentum Ratio (TMR), defined as the ratio of oxidizer momentum to fuel momentum, was varied at 0.85, 1.02, and 1.20. Results showed that TMR had a greater influence on combustion efficiency compared to BF. All tests were conducted at a fixed skip ratio of 0.63, defined as the ratio of skip length (L_s) to pintle diameter (D_p), and the influence of other skip ratios was not explored. Subeom et al. [19] performed a numerical and experimental analysis on the effect of slit heights and circular orifices. Studies concluded that rectangular slits are more efficient in terms of mixing. The study also highlights that in pintle injector design, slits are preferred over simple circular orifices because they form thin propellant sheets, enhancing atomization and mixing. Rectangular slits, in particular, produce a more uniform spray and allow better control over the spray angle, improving combustion efficiency. This geometry also supports enhanced throttling, as the flow distribution can be adjusted more effectively than with circular holes. Furthermore, the injector is designed with two rows of slits, which offers greater operational flexibility. During low thrust operation, propellant can be primarily injected through the lower row, while the upper row becomes more active at higher thrust levels, thereby enabling

smooth throttling across a wide operating range. The staged use of two slit rows also helps maintain efficient atomization even at reduced mass flow rates, ensuring stable combustion under varying operating conditions. Lucchese et.al [20] studied the impact of chemical modelling for LOx/GCH_4 pintle injector on numerical analysis. The study was carried out on multi slit type pintle injector on which three chemical kinetic models were studied. The chemical kinetics namely, JL-9 mechanism, Zhukov 13 species skeletal mechanism (TSR-CDF-13) and Zhukov's detailed scheme 24 species mechanism (TSR-GP-24) were studied. The study highlights the importance of chemical kinetic mechanism by a comparative study. From the studies it was observed that JL-9 mechanism predicts an unphysical flame with high temperature zones at oxidizer rich regions, whereas the skeletal mechanisms more accurately captured the combustion physics by predicting a physically realistic flame structure, heat release distribution, and improved gas-liquid phase interaction, thereby enabling more reliable representation of flame anchoring, mixing, and oxidizer jet penetration inside the combustion chamber.

Many modern LOx-CH_4 engines operate at supercritical pressures, where chamber pressure exceeds the critical points of both propellants. In this regime, the distinction between liquid and gaseous phases disappears, and the mixing process is dominated by diffusion driven mechanisms rather than conventional droplet atomization. While supercritical combustion offers benefits such as enhanced mixing and stable flame anchoring, it also presents modelling and design challenges due to the complex variation of thermo-physical properties near the critical point. Under such conditions, injector geometry becomes the dominant factor in determining how efficiently propellants mix and react. A critical component influencing rocket engine performance is the injector, which governs propellant mixing, combustion efficiency, and stability. Throttling capability is essential for high performance LOx-CH_4 which can be used for reusable launch vehicles and planetary missions, where precise thrust control and deep throttling are required [21]. To meet these requirements, a systematic design and numerical analysis of pintle injectors for LOx–CH_4 engines is essential that provide wide throttleability, stable combustion, and reliable performance for advanced liquid rocket engines.

Our literature review reveals that geometric parameters annular gap [22], skip ratio[23]and pintle geometry[18, 24]play a crucial role in determining combustion performance. This highlights the need to identify an optimal design that can enhance combustion efficiency and stability in oxidizer-centered LOx-methane injectors. The annular gap regulates oxidizer sheet thickness, jet penetration, and momentum exchange between fuel and oxidizer. Pintle slit geometry governs the injection of LOx into the combustor and significantly influences mixing, as the gas-liquid interaction varies with slit geometry. Together, these parameters directly affect flame shape, mixing efficiency, and combustion stability. Despite their importance, relatively few studies have systematically investigated their combined effects, particularly under supercritical LOx-methane conditions.

In this study, a analytical design methodology for a multi slit type pintle injector is developed. An initial baseline configuration is formulated using analytical relations, followed by a detailed optimization of the injector design parameters through CFD analysis. Supercritical combustion simulations of a single element combustor are then performed to evaluate the impact of injector geometric parameters on combustion performance. A baseline

injector configuration is initially designed using analytical model presented. Parametric studies on the annular gap and skip ratio are then conducted using the baseline design to evaluate their influence on mixing and combustion characteristics. Subsequently, the effect of the number of slits is analyzed, followed by a detailed investigation of single row slit configuration. While existing literature highlights the necessity of double row slit arrangements, the detailed parametric analysis presented in this work demonstrates that a single row configuration with increased slit height can achieve similar combustion efficiency. The work establishes a systematic approach for high pressure LOx-methane pintle injectors and examines how these parameters affect mixing, flame characteristics, and overall combustion efficiency, providing clear design guidelines to support the development of efficient LOx-CH_4 propulsion system.

2 Design Philosophy: Pintle Injector

In this study, we design a pintle injector for a typical high pressure operating condition of LOx-methane engine with chamber pressure of 90 bar. The propellant mass flow rates are fixed based on maximum operating pressure of designed single element combustor, and at a mixture ratio of 3.4. Design is carried out for an injector flow rate corresponding to nominal operation of a single element combustor with throat diameter of 38 mm. A two row slit configuration is preferred for improved throttling capability, where the lower row of slits remains active even at low throttling levels, while the effective flow area of the upper row is controlled to achieve throttling[24]. The second row of slits is positioned between the gaps of the first row, which enhances jet interaction and promotes better mixing. Fig.2 shows a schematic of the typical pintle injector, with two rows of oxidizer outlet slits (row-1 and row-2) at the central oxidizer path and an annular fuel gap and the figures provide a representative pintle injector design suitable for high pressure LOx-methane engine. The major pintle design parameters are presented in Table 1for quick reference. The fundamental relations used in the design include the definitions of thrust momentum ratio (Eq. 1), mass flow rate, and blockage factor (Eq. 2), which governs the flow distribution and slot sizing. Using these relations, the annulus area for fuel flow and the slot area for oxidizer flow are calculated.

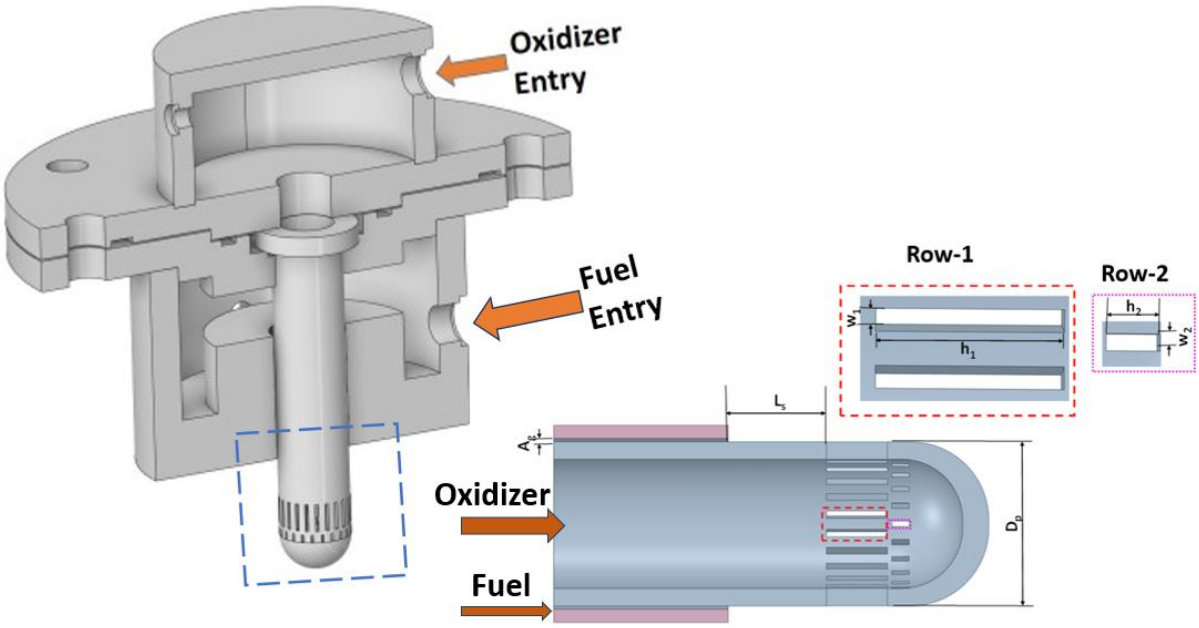

Fig.2 Schematic of pintle injector

Table 1 Design parameters of pintle injector

Symbol	Design Parameter
A_g	Annular gap
D_p	Diameter of Pintle
h_1	Height of Slit in Row-1
h_2	Width of Slit in Row-1
w_1	Height of Slit in Row-2
w_2	Width of Slit in Row-2
L_s	Skip Distance
N	Number of Slits in each row
BF	Blockage Factor

$$TMR = \frac{\dot{m}_o \,.\, \dot{u}_o}{\dot{m}_f \,.\, \dot{u}_f} \tag{1}$$

$$BF = \frac{n.(w_1 + w_2)}{\pi.D_p} \text{ X } 100\% \tag{2}$$

$$A_a = \pi(g^2 + g.D_p) \tag{3}$$

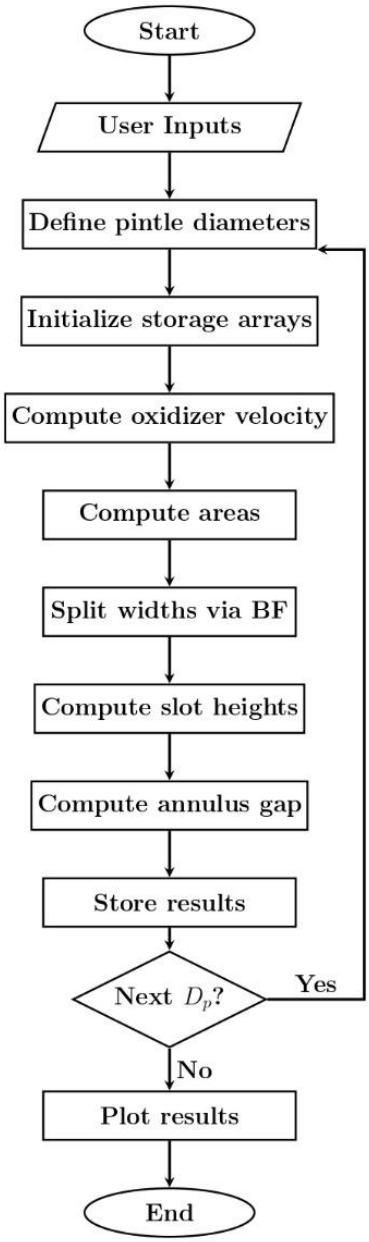

Fig.3 Design algorithm of pintle injector

Fig.3 presents the design algorithm used to design the pintle, implemented through a MATLAB code. Subsequently, the oxidizer flow is split between two rows of slots, and the corresponding slot widths and heights are determined based on the blockage factor and pintle circumference. Finally, the annular fuel gap is obtained from the geometric relation expressed in Eq. 3. Together, these equations form the basis for designing the pintle geometry.

Fig.4 illustrates the variation of slit height with pintle diameter, and Fig.5 shows the corresponding variation of slit width with pintle diameter for three blockage factors (BF = 0.51, 0.58, and 0.65), as obtained from the proposed design methodology. The graphs show that with increase in pintle diameter both the slit heights h_1, h_2 drops, whereas the width of row-1 slits (w1) and width of row-2 (w2) slits increase with increase in pintle diameter. The overall trend of both slit heights and width remains same Literature suggests that higher slit

height is preferred for better mixing [7]. It is observed that increasing pintle diameter reduces the slit height. To maintain higher slit heights, a pintle diameter between 15-20 mm can be considered an optimal choice. For smaller pintle diameters, the slit width becomes very small, resulting in a low width to length ratio, which can complicate the design and make manufacturing more challenging. Based on the mass flow requirements, mixing efficiency, and geometric constraints, a pintle diameter of 18 mm is considered for the initial design configuration. The corresponding design parameters are derived from this selection, ensuring proper propellant distribution and stable combustion. These parameters are summarized in Table 2. Using these optimized values, the initial pintle injector geometry is developed.

Table 2 Dimensions of designed pintle Injector

Initial Design Parameters of the Pintle						
N	h_1 x w_1 (mm)	h_2 x w_2 (mm)	D_p (mm)	Skip Ratio	BF (%)	Annular gap (mm)
22	7 x 0.86	2 x 0.6	18	0.63	58	2.7

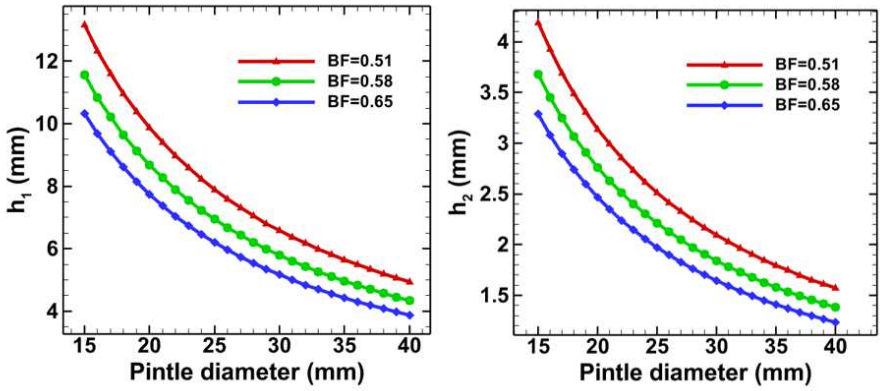

Fig.4 Variation of slit height with D_p at BF: 0.51, 0.58, 0.65

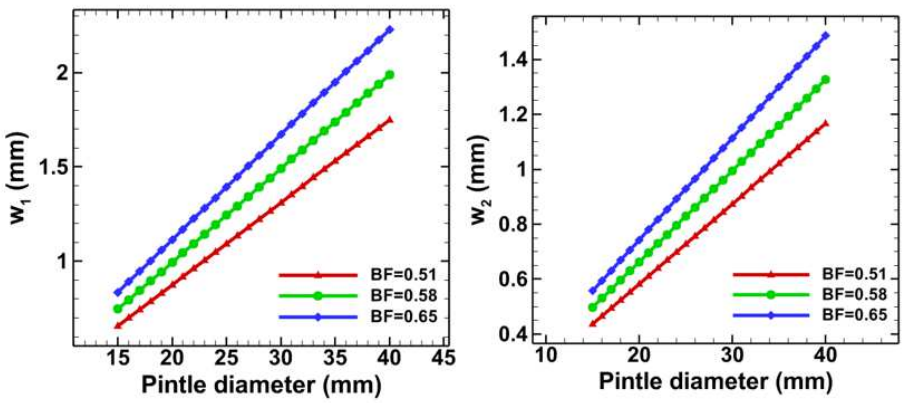

Fig.5 Variation of slit width with D_p at BF: 0.51, 0.58, 0.65

We develop an analytical design methodology and an initial injector configuration is proposed. Based on the analytical design relations and inputs from the literature, a multi row, multi slit pintle injector is configured. This initial design is subjected to numerical analysis, and a detailed optimization is carried out. The objective of the numerical analysis is to

achieve improved mixing efficiency while ensuring the thermal safety of the pintle injector. The following section provide details on optimization studies conducted on base line pintle configuration.

3 Numerical Framework

Rocket engines typically employ non-premixed combustion, where fuel and oxygen are supplied through different injector channels as distinct streams, since this approach is safer and more reliable. Therefore, a non-premixed combustion model is chosen for the simulations, in which the solver treats the fuel and oxidizer as separate streams before diffusion, mixing, and combustion. In this study, flamelet generated manifold (FGM) approach is employed for combustion modelling. In the present work, combustion modelling is performed using a flamelet based approach, following the validated methodology developed by Sharma et al. [25, 26]. The Soave Redlich Kwong (SRK) real-fluid equation of state (EoS) is employed as the thermodynamic model (Eqs.4-7) as it accurately captures the non-ideal behaviour of the propellants, which are injected at subcritical conditions and combust under supercritical conditions, leading to significant variations in density and other thermodynamic properties essential for predicting propellant mixing and combustion performance, together with the Favre-averaged conservation equations for mass, momentum, species, and energy, while accounting for compressibility effects.

$$P = \frac{R.T}{v-b} - \frac{a\alpha}{v(v+b)} \tag{4}$$

$$a = \frac{0.4278\, R^2 Tc^2}{Pc}, b = \frac{0.08664 RTc}{Pc} \tag{5}$$

$$\alpha = \left[1 + k\left(1 - Tr^{\frac{1}{2}}\right)\right]^2, Tr = \frac{T}{Tc} \tag{6}$$

$$k = 0.480 + 1.547w - 0.176w^2 \tag{7}$$

A probability density function (PDF) framework is used to connect turbulence characteristics with chemical reactions by providing pre-tabulated mean values of parameters such as mixture fraction, progress variable, and enthalpy. SST k–ω turbulence model is utilized for turbulence modelling. The governing equations can be referred from earlier work [26, 27].Governing equations are solved in finite volume methodology of ANSYS Fluent®.The Favre averaged governing equations are employed and the equations are given as follows:

- Continuity equation

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_j}\left(\bar{\rho}\tilde{u_j}\right) = 0 \tag{8}$$

- Momentum equation

$$\frac{\partial}{\partial t}(\bar{\rho}\tilde{u_i}) + \frac{\partial}{\partial x_j}\left(\bar{\rho}\tilde{u_i}\tilde{u_j}\right) = -\frac{\partial \bar{P}}{\partial x_i} + \frac{\partial}{\partial x_j}(\overline{\tau_{ij}} - \overline{pu_i''u_j''}) \tag{9}$$

- Energy equation:

$$\frac{\partial}{\partial t}(\bar{\rho}\tilde{h}) + \frac{\partial}{\partial x_j}(\bar{\rho}\tilde{h}\tilde{u}_j) = \frac{\partial \bar{P}}{\partial x_i} + \tilde{u}_j \frac{\partial \bar{P}}{\partial x_j} + \overline{u_j''\frac{\partial \bar{P}}{\partial x_j}} + \frac{\partial}{\partial x_j}\left(\lambda \overline{\frac{\partial T}{\partial x_j}} - \overline{ph''u_j''}\right) + \overline{\tau_{ij}}\frac{\partial \tilde{u}_i}{\partial x_j} + \overline{\tau_{ij}\frac{\partial u_i''}{\partial x_i}} \tag{10}$$

The Favre-averaged transport equation for the mixture fraction $\bar{f}$and un-normalized progress variable are solved in RANS framework. The transport equations are given as:

$$\frac{\partial}{\partial t}(\bar{\rho}\bar{f}) + \nabla.(\bar{\rho}\vec{v}\bar{f}) = \nabla.\left[\frac{\mu_{eff}}{\sigma_t}\nabla.\bar{f}\right] \tag{11}$$

$$\bar{\rho}\frac{\partial \bar{Y}_c}{\partial t} + \frac{\partial}{\partial x_i}(\bar{\rho}\tilde{u}_i\bar{Y}_c) = \frac{\partial}{\partial x_i}\left(\bar{\rho}D_{eff}\frac{\partial \bar{Y}_c}{\partial t}\right) + \bar{S}_c \tag{12}$$

where$\mu_{eff}$is effective viscosity, composed of laminar (μ_l) and turbulent (μ_t) contributions. Within the FGM framework, the progress variable source term $\bar{S}_c$is represented using a finite-rate reaction model derived from the flamelet database. The mixture fraction $\bar{f}$and the progress variable c are used as the governing parameters to extract this source term. Therefore,$\bar{S}_c$ determines the turbulent flame behaviour and its location, and is calculated as:

$$\bar{S}_c = \bar{\rho}\iint S_{FR}(c,f)\, p(c,f)\, dc\, df \tag{13}$$

4 Computational Domain

A single element combustion chamber is designed to conduct parametric analysis on designed pintle configuration. The chamber is developed using an in-house MATLAB code [28] and configured to accommodate a single pintle injector element. The input parameters for the combustor design are the oxygen to fuel mass flow ratio and the chamber pressure. Fig.6 shows the 2D schematic of the combustor, with dimensions normalized by the throat diameter.Pintle injector configurations are generally observed to employ longer combustion chamber lengths to facilitate effective propellant mixing, flame development, and complete, stable combustion. Accordingly, to adequately capture flame development and provide sufficient residence time for combustion, a combustion chamber with a characteristic length of 1.3 m is adopted in the present study.

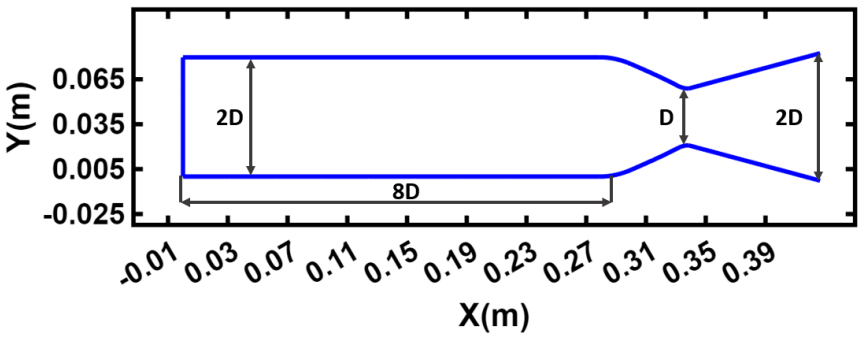

Fig.6 Schematic of designed combustor

Fig.7illustrates the 3D schematic of the combustor, where a protruding pintle is clearly visible. A 90° sector of the fluid domain is extracted from the full combustor geometry for the computational analysis. It is shown adjacent to the three dimensional model. The combustor is truncated downstream of the throat, while retaining a minimal divergent nozzle section. The truncation is justified by the fact that the flow reaches sonic conditions (Mach 1)

at the throat, resulting in choked flow. A minimal divergent section is retained downstream of the throat to ensure a smooth pressure transition at the outlet. In the meshed computational domain shown, methane is injected through the annular gap surrounding the pintle, while liquid oxygen (LOx) is introduced through the central region of the pintle. The combustor walls are assumed to be adiabatic, implying negligible heat transfer to the surroundings. The outlet of the domain is specified as a pressure outlet to allow for proper exhaust flow boundary conditions.

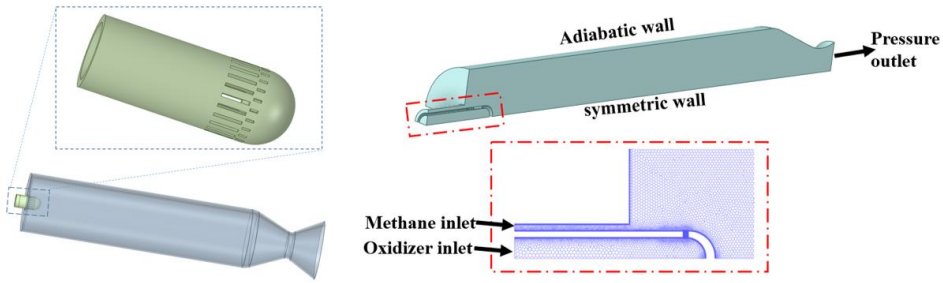

Fig.7 Computational domain and protruded pintle in the combustor segment

The governing equations for mass, momentum, energy, and transport are solved using a pressure based formulation, with pressure velocity coupling handled through the SIMPLE algorithm. Convergence is evaluated based on the reduction of scaled residuals below 10^{-3} for all solved equations. In addition to residual convergence, key integral flow variables such as chamber pressure and temperature, are monitored to ensure solution stability and the attainment of a converged solution.

5 Grid Independence

Grid independence study is performed to ensure accurate prediction of combustion and mixing performance. Three mesh densities are evaluated; a coarse mesh with 768,660 elements, a medium mesh with 1,394,577 elements, and a fine mesh with 2,263,247 elements. All meshes are generated with a body of influence extending 100 mm from the combustor inlet, to adequately resolve the initial mixing region and capture early stage flame development. Boundary layer meshes are applied along all walls to accurately resolve near wall gradients of velocity, temperature which are critical for predicting wall heat flux and local combustion characteristics. Polyhedral cells are employed due to their superior convergence behaviour, flexibility in capturing complex geometries, and efficient resolution of flow gradients in reacting flows. The analysis is performed with single element mass flow rates of 4.35 kg/s for LOx and 1.27 kg/s for CH_4, at injection temperatures of 110 K and 250 K, respectively. Figure 8 show the results of the grid independence study. Fig.8(a) shows the variation of density along the combustor central line along the axis, indicating minimal differences between all three meshes and a consistent trend across the domain. Fig.8(b) presents the temperature distribution for the three mesh densities. In the body of influence region (first 100 mm of the combustor), all meshes display a similar trend with negligible variation. Beyond this region, the coarse mesh (≈768,660 elements) predicts slightly higher temperatures, likely due to its larger cell size and increased numerical diffusion. In contrast, the medium and fine meshes exhibit nearly identical temperature profiles with only minor deviations. Fig.8(c) shows the oxygen mass fraction along the central line. Similarly, in the body of influence region, all meshes display comparable profiles while beyond this region, the coarse mesh predicts slightly lower oxygen mass fractions, whereas the medium and fine

meshes remain nearly identical, indicating convergence of the solution. Based on these observations, the medium mesh is selected for all subsequent simulations, as it provides a converged solution while maintaining computational efficiency.

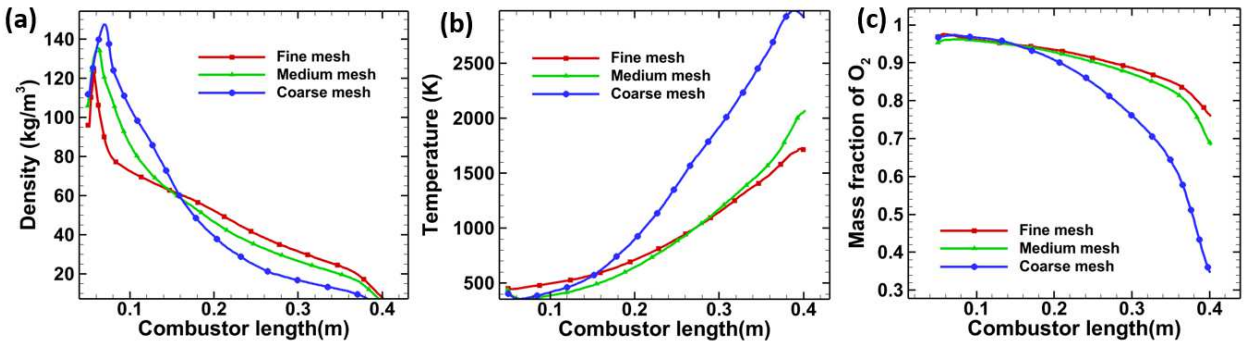

Fig.8 Grid comparison plots (a) density, (b) temperature and (c) mass fraction of oxygen at central axis line

This setup enables a detailed analysis of flow mixing, combustion, and heat release characteristics within the single element combustor.

6 Results & Discussion

6.1 Analysis of initial design

Results of design configuration tabulated in Table 2 with an annular gap of 2.7 mm and a skip ratio of 0.63 are presented in this section. A detailed explanation is provided on the flame structure, mass fractions of the propellants, and the combustion products.

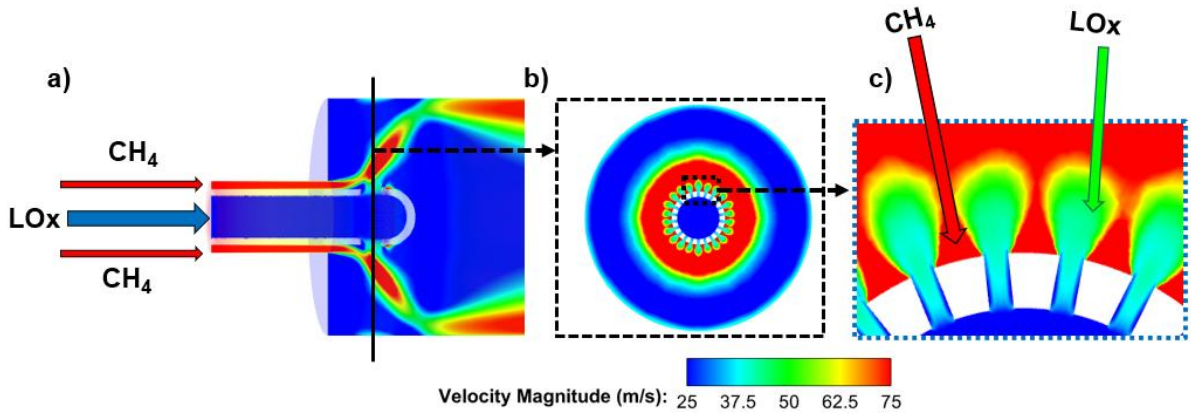

Fig.9 Velocity Magnitude (m/s) (a) entry region (b) radial plane at x=0.036m (c) row-1 slit exit

Fig.9(a) shows the velocity magnitude at entry region of pintle, CH_4 enters through annular region at a velocity of 75 m/s and LOx enters from the central pintle at a velocity of 25 m/s. Fig.9(b) shows the radial velocity magnitude contour near the pintle depicting the flow through the pintle and annular gap. Fig.9(c) shows LOx entering through the central pintle and exits via multi slit holes, impinging on the annular methane flow. The designed annular gap provides a velocity ratio of 3 that is required for shear dependent mixing. Higher velocity ratio is resultant of high density ratio used in the configuration. This pintle configuration is different from pintle operating at sub-critical operating condition, which operates at low density ratio of 0.4, the mechanism of atomization is clearly different from super critical pintle injector operation.

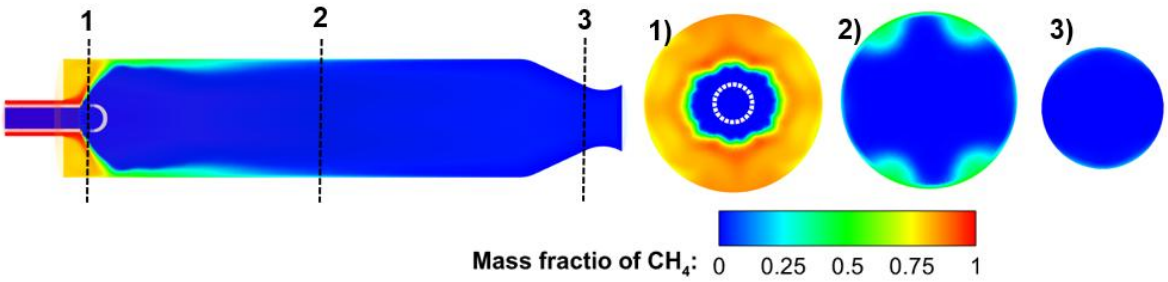

Fig.10 Mass fraction of CH_4 at central axial and radial planes of initial design

Fig.10depicts the mass fraction contours of CH_4. Methane with mass fraction 1 enters into the chamber through the annular gap region and diffuses along the combustor walls due to radial exit of high flow rate oxygen stream. The reduction of methane mass fraction indicates methane consumption along the length of combustor, with no methane present at the end of combustor. Fig.11 shows the iso-surface of oxygen representing oxygen entry through the slits. Fig. ***12***shows the mass fraction of oxygen on the central axial plane and selected radial planes. LOx enters the combustor through the central pintle. The first radial plane illustrates the oxygen mass fraction as it enters the combustor through the slits. The contours on the downstream radial planes show progressive oxygen consumption along the length of the combustor. However, a substantial amount of oxygen remains unconsumed at the nozzle throat. Analysis of this injector configuration highlights unconsumed LOx leaving the combustor, which can lead to a lower chamber pressure rise.

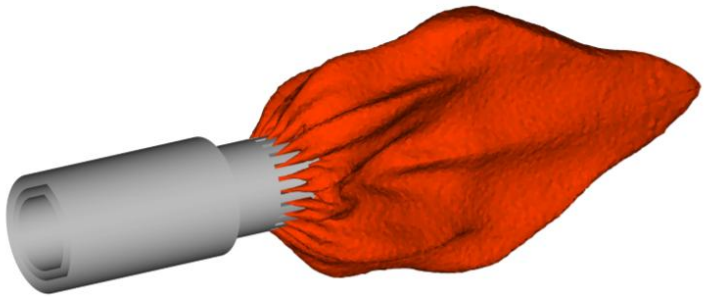

Fig.11 Iso surface of oxygen

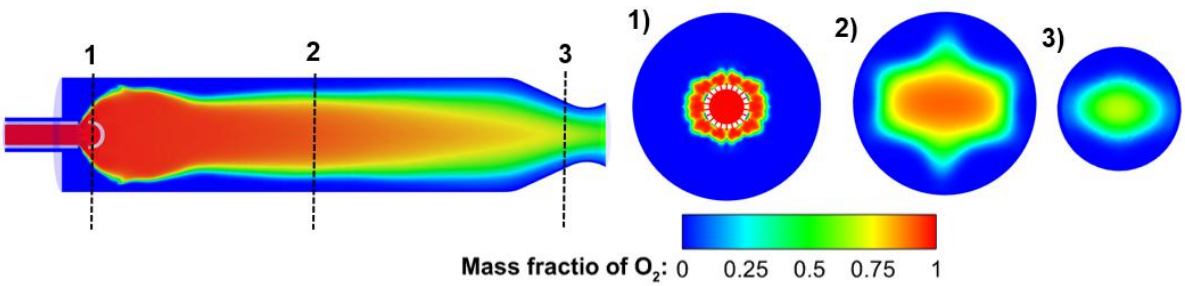

Fig. 12 Mass fraction of LOx at central axial and radial planes of initial design

Fig.13shows the temperature contour, with a maximum flame temperature of 3600 K and the flame anchored at the pintle post. The peak temperature is observed along the region of stoichiometric mixture ratio. The temperature rise near the pintle slits indicates instantaneous mixing of the propellants, whereas the high momentum of the oxidizer jet pushes the flame downstream, inhibiting homogeneous mixing in this configuration. The

lower mixing is observed because of the lower interaction between the oxidizer and the fuel streams.

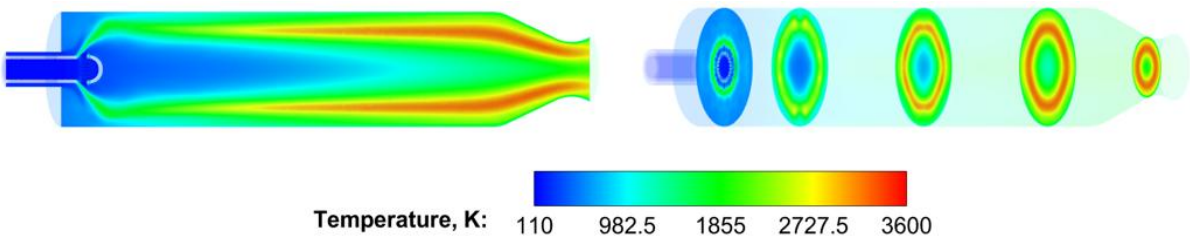

Fig.13 Temperature contour of initial design at radial and central axial plane

Fig.14 depicts the mass fraction of H_2O and CO_2 at axial cut planes. The maximum mass fraction of H_2O and CO_2 is observed to be similar to temperature profile depicting the combustion is happening at the stoichiometric mixture ratio region. The mass fraction is around 0.29 for CO_2 and 0.4 for H_2O signifies that other intermediate product (CO, H_2, OH) are present near the throat region. Mass fraction contours also indicate inhibited mixing in this injector configuration. The uniformity index, represents how evenly or uniformly a parameter is distributed across a surface of temperature at the exit is observed to be 0.92.

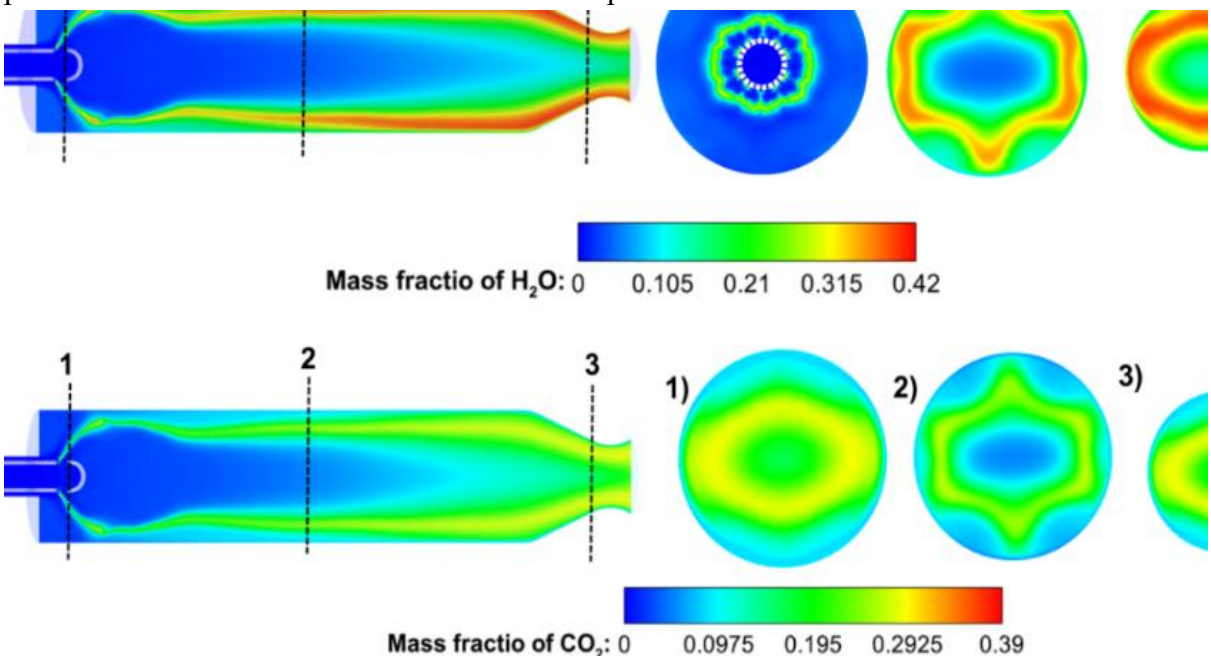

Fig.14 Mass fractions of CO_2, and H_2O of initial design

Analysis of the initial design configuration shows that mixing is limited, with a substantial amount of oxygen with maximum mass fraction of 0.59 concentrated near the throat. This indicates inadequate mixing and lower overall oxygen consumption. The results suggest that the initial configuration can be improved through parametric analysis of the pintle design. In following section, a systematic exploration of major design parameters of pintle injector has been conducted.

6.2 Impact of annular gap

To optimize the design, the impact of annular gap is studied for the initial design at 2.7mm, 2.3mm, and 2mm annular gap and to investigate the sensitivity of design parameters

to velocity ratios for TMR values of 1.1, 0.94, and 0.81 respectively. Table 3shows the design modifications considered for the study, where case-ID refers to initial design, G1 and G2 are the two annular gaps simulated. The objective of this investigation is to examine how reducing the annular gap area affects the injector performance, with the expectation that a smaller annular gap would increase the fuel injection velocity and thereby enhance the mixing process and the results are discussed in this section.

Table 3 Simulation conditions with varying annular gap

Cases	N	h_1 x w_1 (mm)	h_2 x w_2 (mm)	U_o (m/s)	U_f (m/s)	TMR	A_g (mm)
Case-ID					75	1.1	2.7
Case-G1	22	7 x 0.86	2 x 0.6	25	90	0.94	2.3
Case-G2					105	0.81	2

Fig.15 presents the injection velocity distributions for annular gaps of 2.7 mm, 2.3 mm, and 2.0 mm. It is observed that the injection velocity increases with a decrease in annular gap. The velocity contours confirm that the expected injection velocities are attained, with methane injection velocities of approximately 75 m/s, 90 m/s, and 105 m/s, respectively.

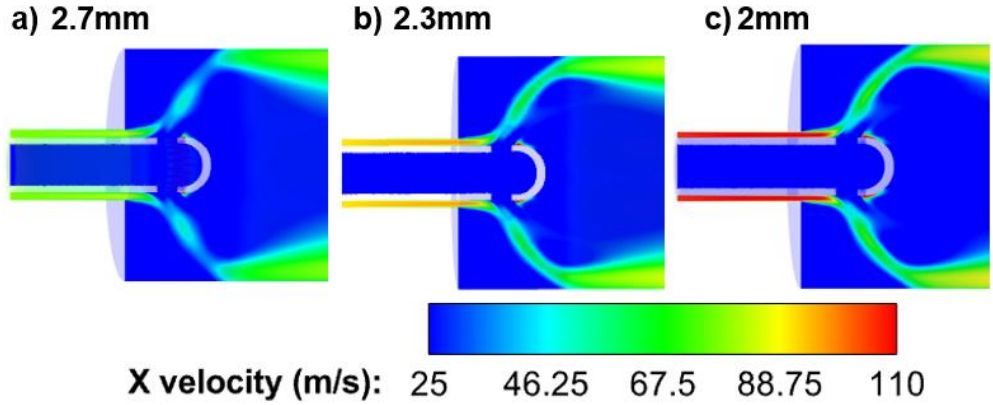

Fig.15 Velocity contour for annular gap 2.7mm, 2.3mm and 2mm

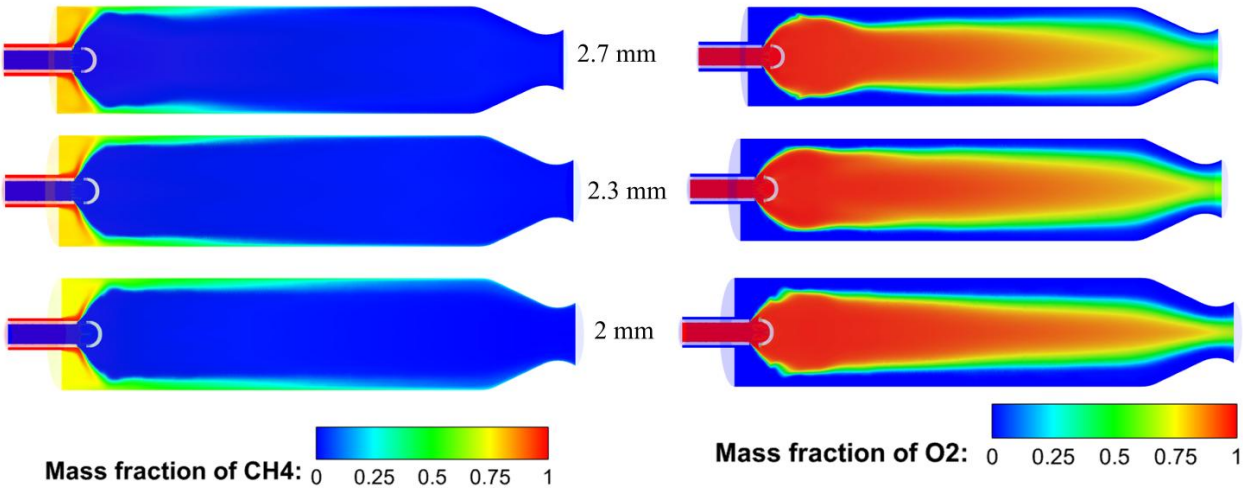

Fig.16 Mass fraction contours of Methane and Oxygen at annular gaps 2.7mm, 2.3mm, and 2 mm

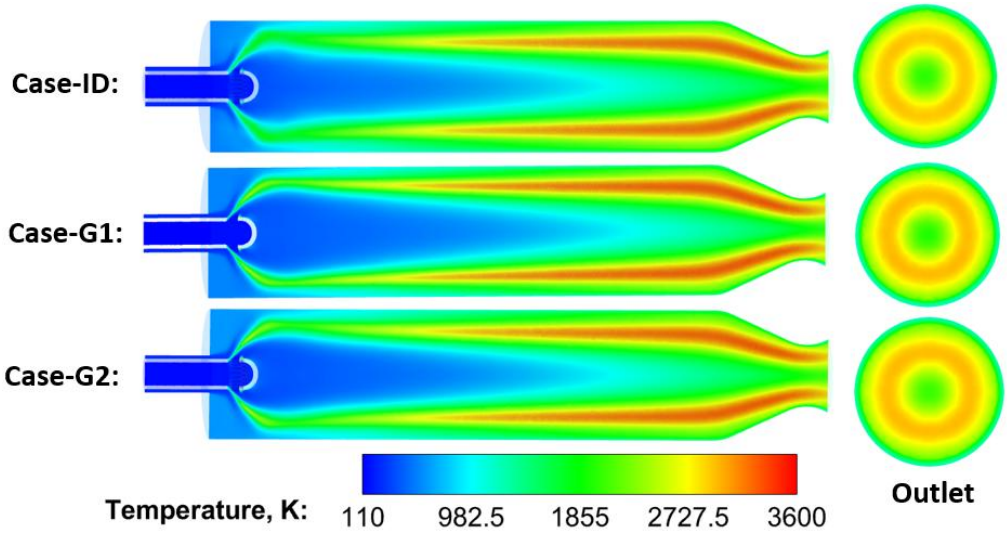

Fig.17 Temperature contour for annular gap2.7mm, 2.3mm, and 2mm

Fig.16 shows the mass contour of propellants CH_4 and LOx. Through the contours it is observed that there is no significant improvement in mixing with variation in the annular gap. Fig.17 shows the temperature contours for different annular gaps in the initial design. The contours indicate that variations in the annular gap do not significantly alter the flame structure or the overall temperature distribution. This occurs as the gas-liquid interaction region is mainly influenced by injector geometry and flow momentum, rather than the size of the annular gap. As a result, changes in the annular gap have no substantial effect on mixing and combustion performance. Based on these observations, the influence of the skip ratio has been investigated and is discussed in the next section.

6.3 Impact of skip ratio

Skip ratio is defined as the ratio of L_s to the D_p. It affects the distance methane travels before interacting with LOx and can influence mixing. To investigate the effect, the initial design is modified by increasing the skip ratio to 1. For this configuration, three annular gaps 2.7mm, 2.3mm and 2mm are examined to evaluate the effect of annular gap and skip ratio on mixing performance. This study provides a clear understanding of the combined effects of skip ratio and annular gap on mixing performance.

Fig.18 shows the comparison between the skip ratios 0.63, 1 for an annular gap of 2mm.The results indicate that there is no significant improvement in mixing with an increase in skip ratio.

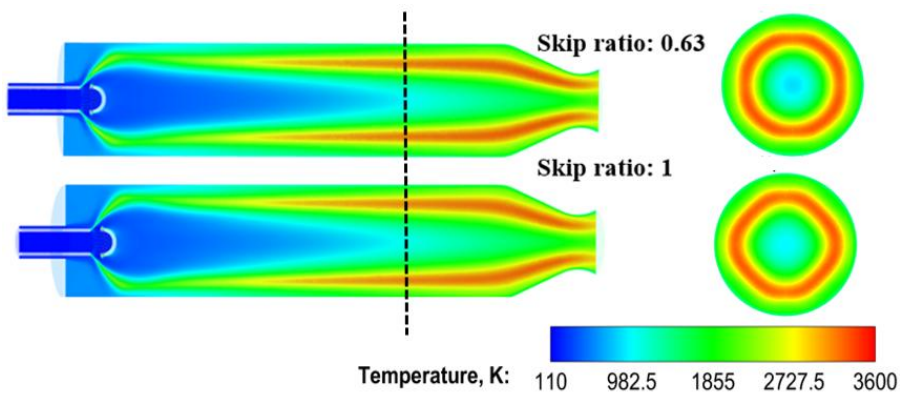

Fig.18 Temperature contour of 2mm annular gap for skip ratio 0.63, and 1.

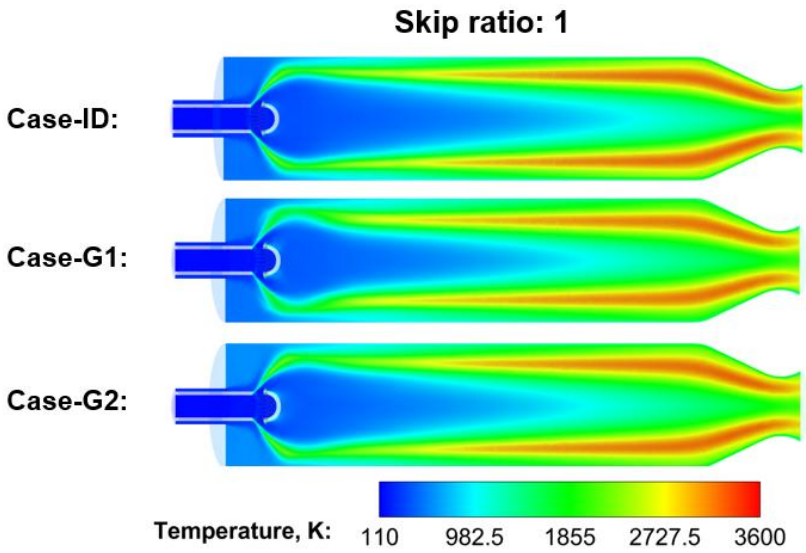

Fig.19 Temperature contour for skip ratio 1 at annular gap 2.7mm, 2.3mm, and 2mm

To study the influence of annular gap at higher skip ratios for skip ratio 1, three annular gaps of 2.7 mm, 2.3 mm, and 2.0 mm are analysed. Fig.19 presents the temperature contours for different annular gaps at a skip ratio of 1.0. The results indicate that varying the annular gap does not lead to improved mixing even at the higher skip ratio. For both the skip ratios, changes in annular gap do not produce significant results on temperature distribution.

Based on the analysis in this section, it can be concluded that neither skip ratio nor annular gap significantly improve mixing. The incomplete flame structure and the high residual oxygen mass fraction near the throat indicate that the injector design requires further optimization to enhance mixing and combustion performance. This suggests that the effect of methane injection on mixing performance is relatively low in this configuration. To further optimize the injector and to understand the parameter that controls mixing, the influence of pintle slit configuration, which governs oxidizer injection, is investigated and discussed in next section.

6.4 Impact of slit geometry

6.4.1 Impact of number of slits

An inhibited propellant mixing is observed in the initial design, which can lead to lower combustion efficiency. Literature highlights [7] the importance of higher slit heights and suggests that increasing slit height can enhance mixing. To achieve higher slit heights while maintaining a same TMR, velocity ratio as per initial design, the numbers of slits in both rows are reduced. A parametric study is performed on the reduced slit counts pintle configurations. Starting from the baseline design with 22 slits, the numbers of pintle slits are reduced to 20, 18 and 16 while utilizing the same skip ratio of 0.63 and an annular gap of 2.7 mm as of initial design. Table 4lists the design parameters considered for this study and the results are discussed in this section.

Table 4 Pintle injector configuration with varying number of slits

Slit configuration	N	h_1 x w_1	h_2 x w_2

		(mm)	(mm)
Configuration-1	22	7x0.86	2x0.6
Configuration-2	20	8x0.85	3x0.40
Configuration-3	18	9.1x0.85	3x0.40
Configuration-4	16	10.4x0.85	3x0.40

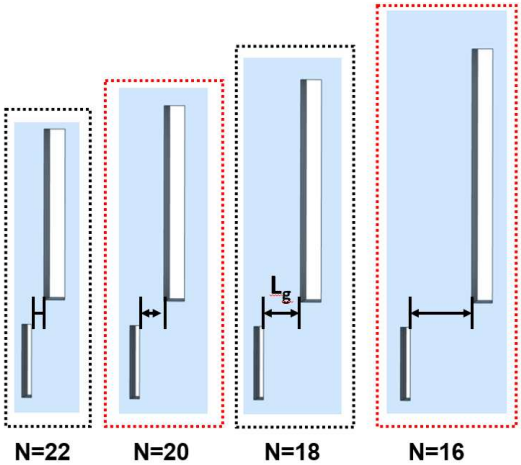

Fig.20 Variation in slit geometry for number of slits 22, 20, 18 and 16.

Fig.20 represents the variation in the slit configuration for the number of slits 22, 20, 18 and 16 respectively. It is clearly observed that the row-1 slit height increases as the number of slits decreases and the gap (L_g) between the slits of row-1 and row-2 increases with decrease in number of slits.

Fig.21 presents the mass fraction of methane and oxygen for varying number of slits. Improved mixing is further supported by the mass fraction contours of oxygen and methane. It is observed the length of the oxygen rich core decreases as the number of slits is reduced. This trend indicates that the modified slit geometry promotes more effective interaction between the oxidizer and fuel streams, thereby improving mixing performance.

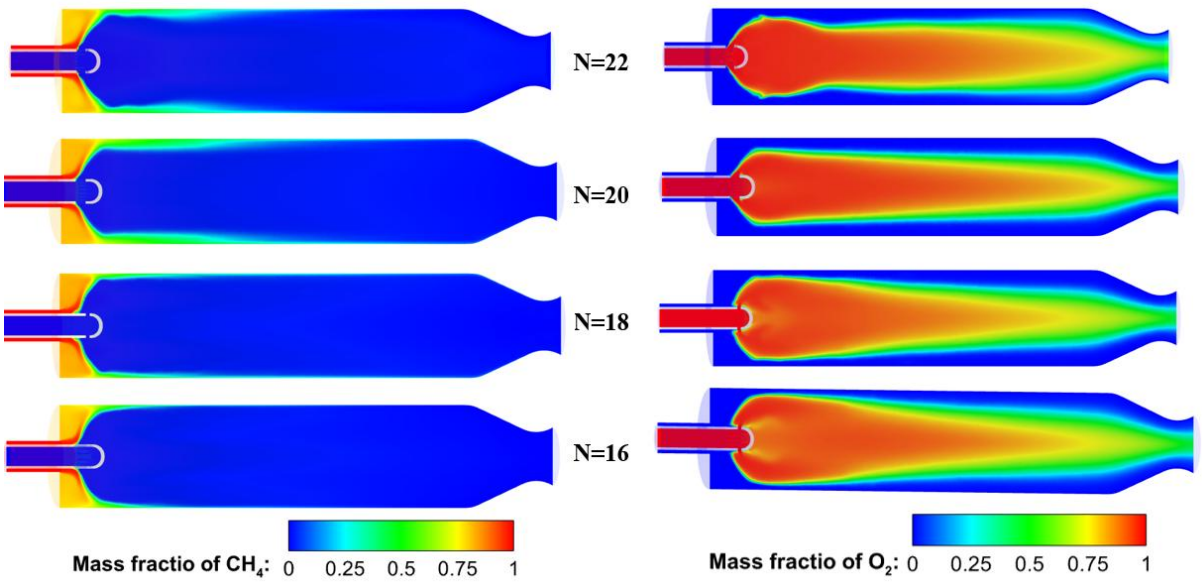

Fig.21 Mass fractions of CH4, Lox at an axial centre plane for different number of slits

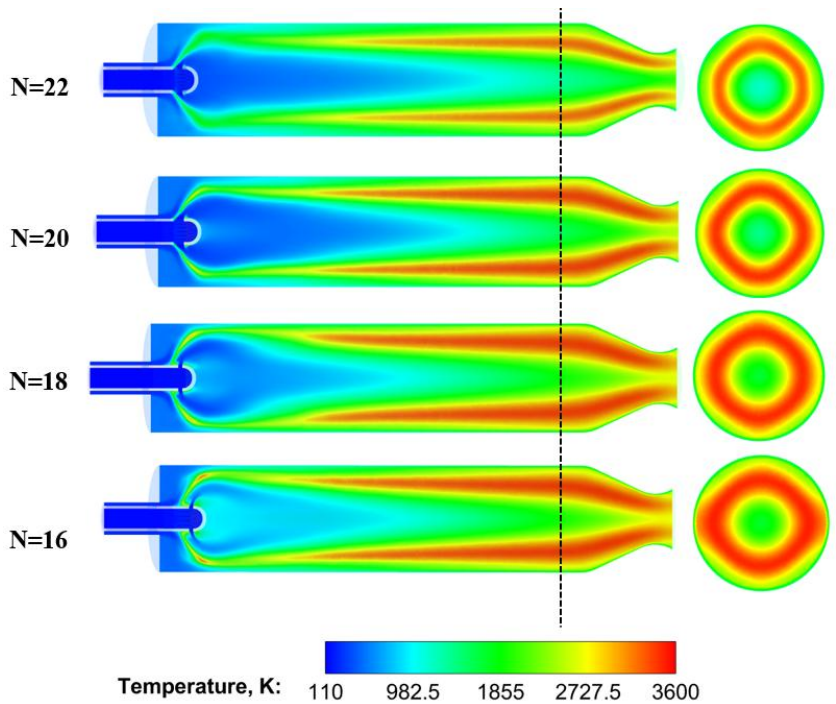

Fig. 22 Temperature contour for different number of slits and temperature plot along the central line

Fig. ***22*** presents the temperature contours for slit configurations of 22, 20, 18, and 16, along with contours at a radial plane of 0.3 m, it is observed that at N=16 the pintle tip temperature is high compared to the other configurations. The results show that the flame width increases as the number of slits is reduced, due to the corresponding increase in slit height. This broader flame zone indicates improved mixing and enhanced combustion performance. In addition, the reduction in low temperature regions suggests the formation of a more stable and uniform flame structure. From the temperature contours it is observed that there is a high temperature zone at pintle tip for N=16. This occurs because reducing the number of slits increases the gap between adjacent slits.

Fig.23 shows the plots along the central line of outlet, Fig.23(a) shows the oxygen mass fraction along outlet central line, it is observed that the higher number of slits more oxygen is observed at the outlet depicting improper mixing and the decrease in oxygen mass fraction reveals the increase consumption of oxygen with decrease in slits, Fig.23(b) shows the CO_2 mass fraction, Fig.23(c) shows the H_2O mass fraction, the increase in mass fraction of bi-products with decrease in slit number further support the finding that decrease in slit number with increase in height resulted in better mixing.

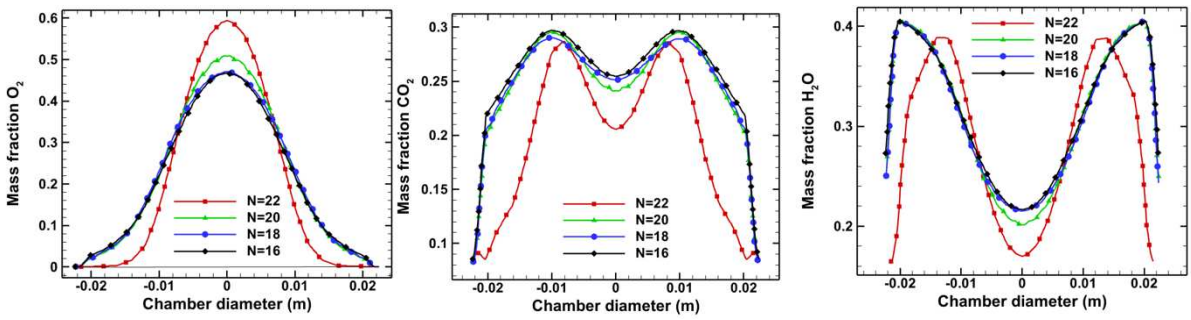

Fig.23 Mass fraction of (a) O_2, (b) CO_2, and (c) H_2O at the central line on outlet

Increase in slit height increase the interaction of the methane stream with LOx more effectively. Fig.24 shows the schematic of increase in jet interaction with increase in slit height. Additionally, the temperature near the pintle tip also increases with a reduction in the number of slits, indicating improved mixing near injector. Although the combustion and mixing performance is high at 16 number of slits, this comes with a disadvantage of high pintle temperature.

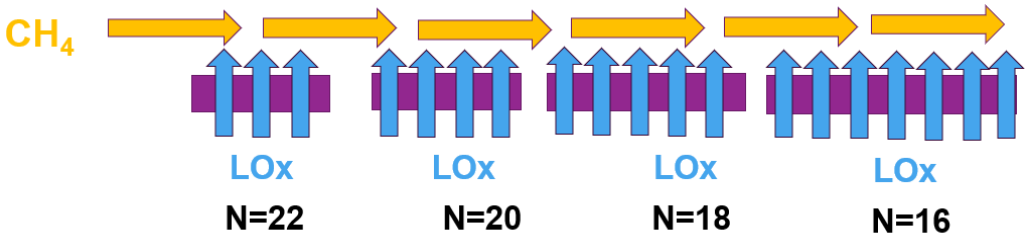

Fig.24 Methane and LOx stream interaction at different slit number

Fig.25 shows the temperature distribution along the central plane starting from the pintle tip. It is observed that the temperature along the combustor length increases more rapidly as the number of slits decreases. The increase in slit height increased the mixing efficiency, as it enhances the interaction between the LOx jets and the methane stream. Fig.26 illustrates the methane flow pattern. Fig.27 shows the methane mass fraction on the pintle injector walls. It is observed that the increase in the gap (L_g) allows methane to bypass and to flow closer to the pintle surface. Since the dimensions of the second row of slits remain unchanged despite the increased gap, methane enters near the pintle tip, and subsequent mixing in this region leads to the formation of a localized high temperature zone.

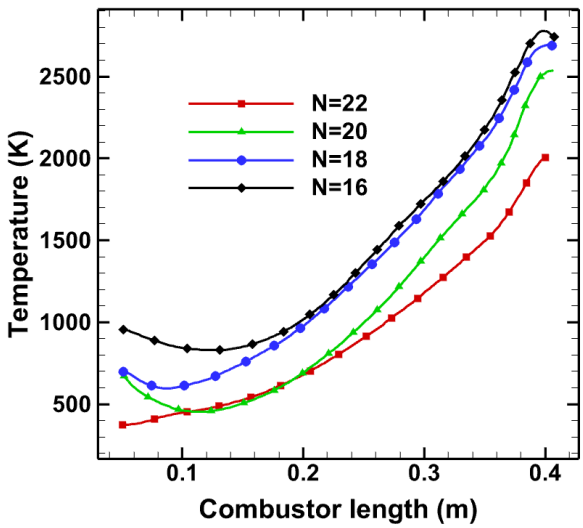

Fig.25 Temperature plot along the central plane starting from the pintle tip for different slits count

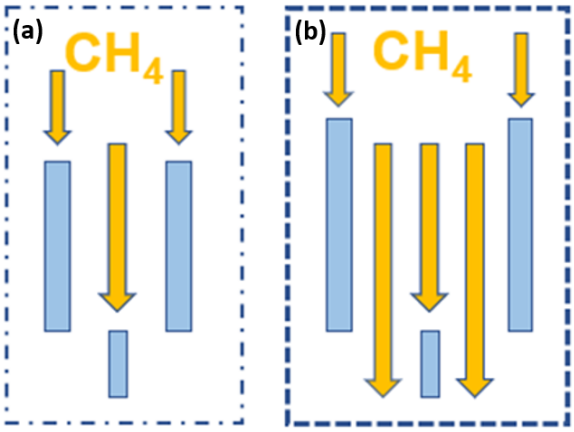

Fig.26 Schematic of Methane flow at a) higher slit count and b) lower slit count

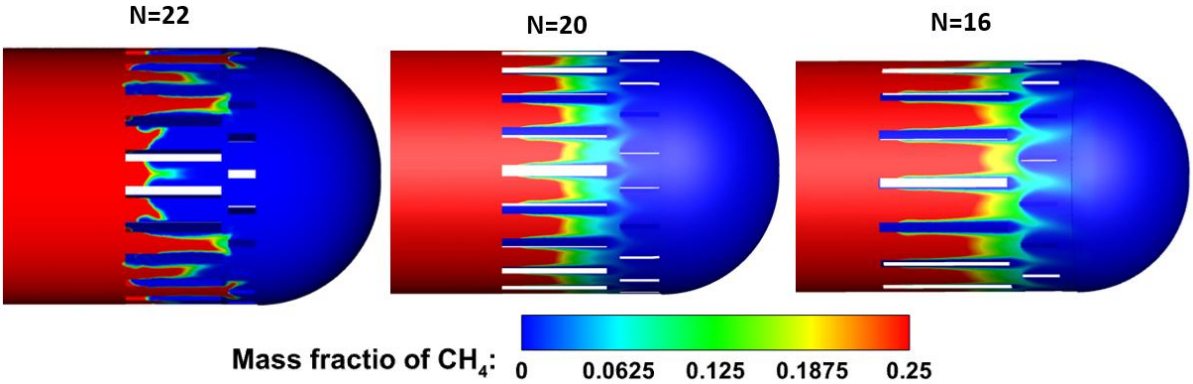

Fig.27 Mass fraction of methane on pintle for a) N=22, b) N=20, c) N=16

This study on impact of number of slit shows that, the configuration with 18 slits, improved mixing is achieved without the formation of a high temperature zone near the pintle tip. This eliminates the need for additional thermal cooling, thereby simplifying the design, increasing pintle life, and improving resistance to thermal stresses. Hence, the N = 18 configuration is identified as the optimum design, offering enhanced mixing performance while maintaining acceptable pintle temperatures. Overall, the improvement in mixing efficiency is achieved primarily through variation in slit number, with minimal influence from annular gap and skip ratio.

From the section, it is observed that the injection parameters governing oxidizer injection plays a dominant role in controlling the mixing behaviour for the present design. Analysis shows that, mixing is weaker at higher number of slits and reduction in number of slits strengthens the mixing performance. However, further reduction in slit count resulted in high temperature zone at the pintle tip. So, the slit count must be optimized to achieve improved mixing while maintain a low temperature zone near the pintle tip.

6.5 Analysis of single slit configuration

In the previous section, the comparison of N= 22, 20, 18, and 16 is performed for double row slit configuration. In this section, we analyse the impact of single row slits in comparison to double row slits designed based on the analytical survey. The analysis is carried out on single row slit configurations, and this section expands it by comparing the entire four configuration N= 22, 20, 18, and 16 for single and double row slit configuration. Figure shows the slit configuration of single row slit and double row slit. The single row slit design is obtained by removing lower row of slits from the double row configuration and

increasing the height of each slit in the remaining upper row, while maintain the same slit width. This modification ensures that the LOx injection area remains same for all cases, thereby preserving the TMR across all the cases. The annular gap is maintained at 2.7mm and skip ratio is 0.63 same as the initial design. The design parameters of single row slits are tabulated in Table 5.

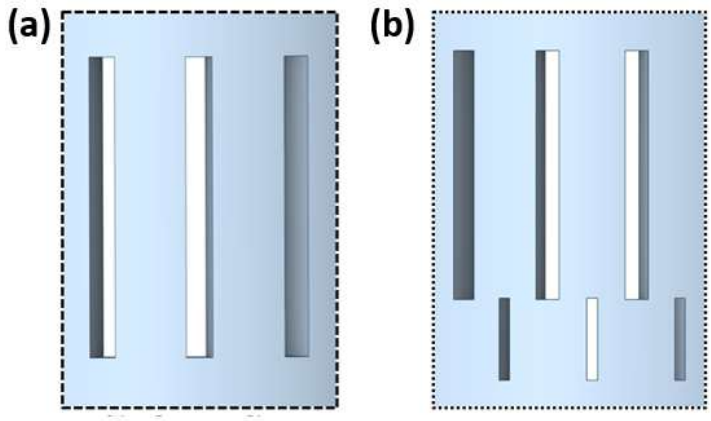

Fig.28 Schematic of a) single row, b) double row slits

Table 5 Design Parameters for single row slit

Configuration	N	h x w (mm)
Configuration-1	22	8.58 x 0.85
Configuration-2	20	9.44 x 0.85
Configuration-3	18	10.5 x 0.85
Configuration-4	16	11.8 x 0.85

Fig. ***29*** shows the comparison of oxygen mass fraction for double row and single row of slits at a central axial plane at N=22, 20, 18 and 16. The high concentration of oxygen as central core is observed to decrease with decrease in number of slits for the both double row and single row of slits. For N = 16, it is observed that near the pintle region the oxygen mass fraction in the single row slit is lower than that of the double row slit, which represents that combustion occurs at a higher rate in this region for the single row slit. Hence, a high temperature zone is observed in the single row configuration. It is also observed that at a lower number of slits, the double row configuration reduces the challenge of a high temperature zone near the pintle tip, as the second row of slits provides secondary mixing and restricts the methane flow toward the pintle.

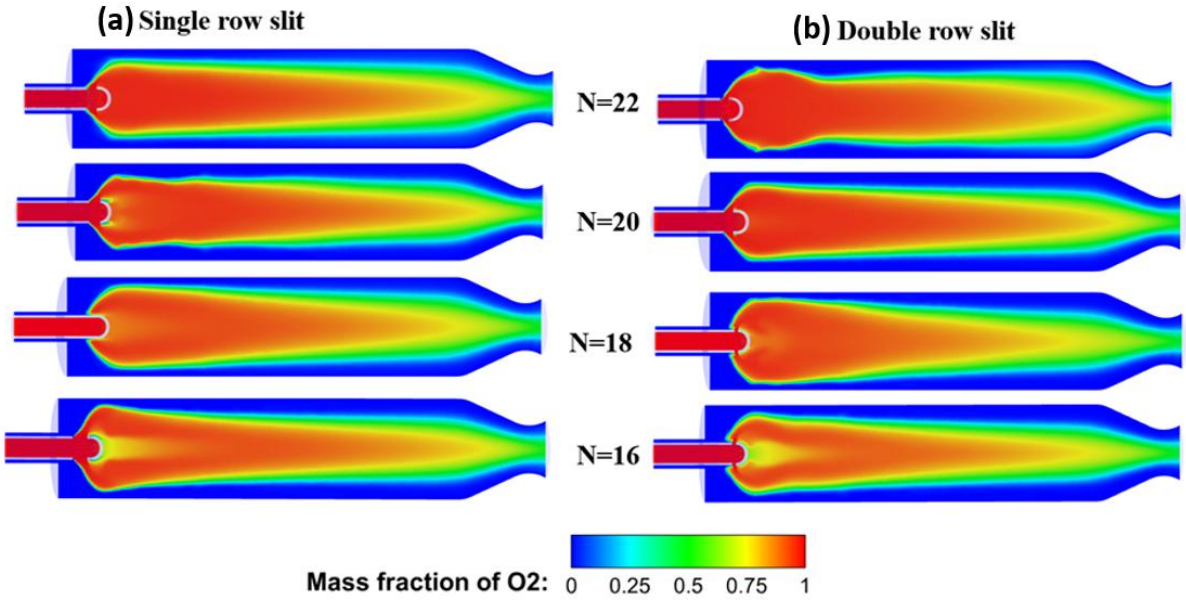

Fig. 29 Mass fraction of CH4 at an axial centre plane of a) single row and b) double row of slits at N=22, 20, 18 and 16

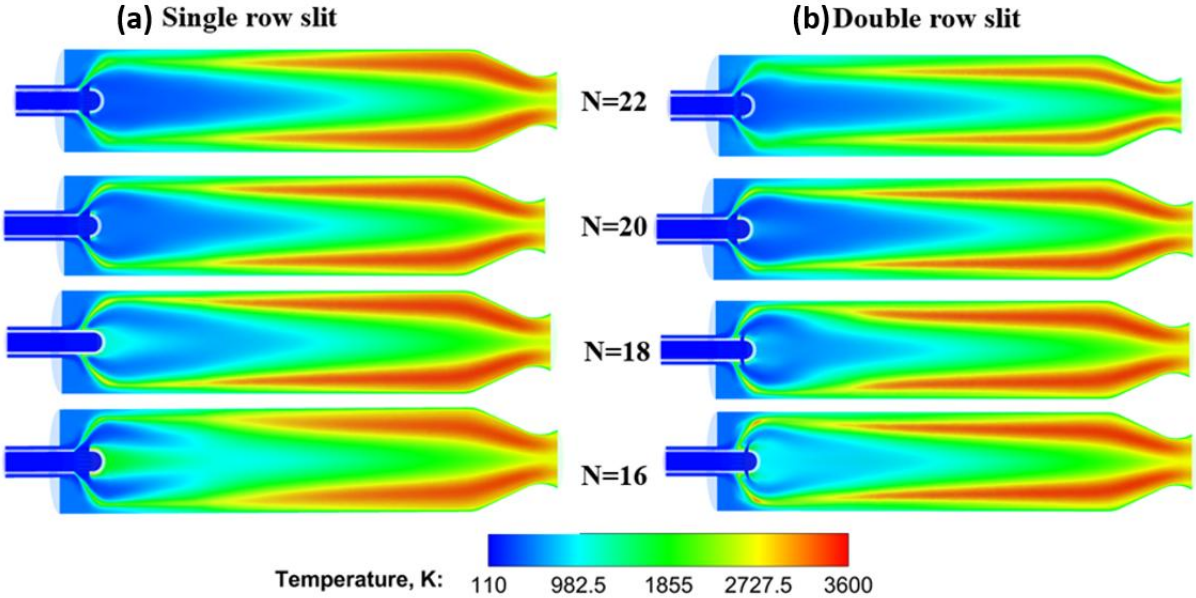

Fig.30 Temperature contour of a) single row and b) double row of slits at N=22,20, 18 and 16

Fig.30 presents the temperature contours comparing single row and double row slit configurations for four cases (N = 22, 20, 18, and 16) at an annular gap of 2.7 mm and a skip ratio of 0.63. For the single row slit configuration, the mixing at N = 22 is higher than that of the corresponding double row slit case. This occurs because the LOx jet interacts more strongly with the methane annular flow. The increased slit height in the single row configuration enhances this interaction, leading to improved mixing. As the number of slits decreases, the mixing improves further in the single row configuration, following the same trend observed for the double row slit configuration. This also results in a high temperature zone near the pintle, which is caused by increased methane bypassing. Figure shows the temperature contours at a radial plane of 0.38m for single row and double row of slits. At higher slit number (N=22,20) the temperature in single row slits is higher compared to double

row of slits because as the L_g is smaller at this case, the lower slits are less active and further increase in height of slits in single improved the mixing. At lower number of slits (N=18,16) the temperature at radial plane 0.38 is higher for double row, as the L_g increases, the lower row of slits (row-2) become more active and the mixing is improved.

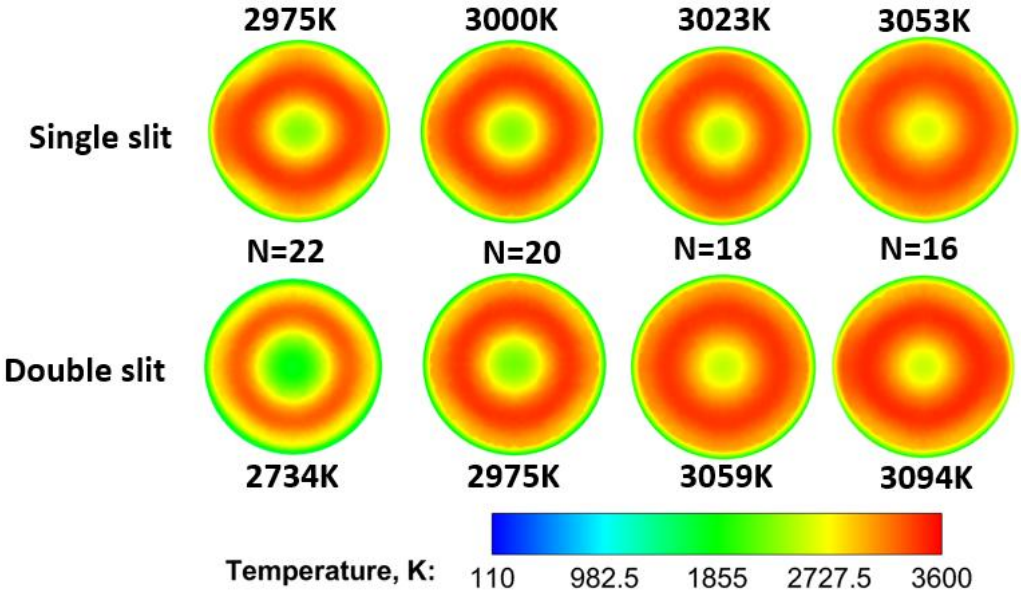

Fig.31 Temperature contours of a) single row and b) double row of slits at N=22, 20, 18, 16 at radial plane 0.38m

The temperature uniformity index at the outlet for the double row slit configuration is found to be 0.922, 0.938, 0.948, and 0.948 for N = 22, 20, 18, and 16, respectively. For the single row slit configuration, the corresponding values are 0.941, 0.9405, 0.942, and 0.9451 for N = 22, 20, 18, and 16, respectively. From the temperature uniformity index, it can be observed that for a higher number of slits, the single row slit configuration exhibits better uniformity, whereas for a lower number of slits, the double row slit configuration provides better temperature uniformity.

From this section, it is observed that the jet interaction area plays a crucial role in mixing. For the N = 22 single row configuration, the interaction area between the LOx jets and the annular methane stream is larger than that of the corresponding double row configuration, which results in better mixing. In addition, since the number of slits is high at N = 22, the methane bypass ratio is low, and therefore no high temperature zone is observed near the pintle tip. Hence, N=22 single slit configuration can be considered as an optimum choice for better mixing and low pintle tip temperature.

6.5.1 Mixing mechanism

The mixing process is predominantly governed by the oxidizer stream, with the oxidizer injection area playing a major role in determining the overall mixing performance and is strongly influenced by the slit configuration, making slit geometry a key design parameter.

Fig.32 illustrates the schematic of the mixing mechanism for the double row and single row slit configurations. The slit height plays a crucial role in enhancing mixing, as longer slits promote stronger interaction between the LOx jets and the methane stream. In the double row configuration, the first row of slits is primarily responsible for the initial mixing

of the propellants, while the residual methane bypassing the first row interacts with the LOx from the second row, resulting in secondary mixing.

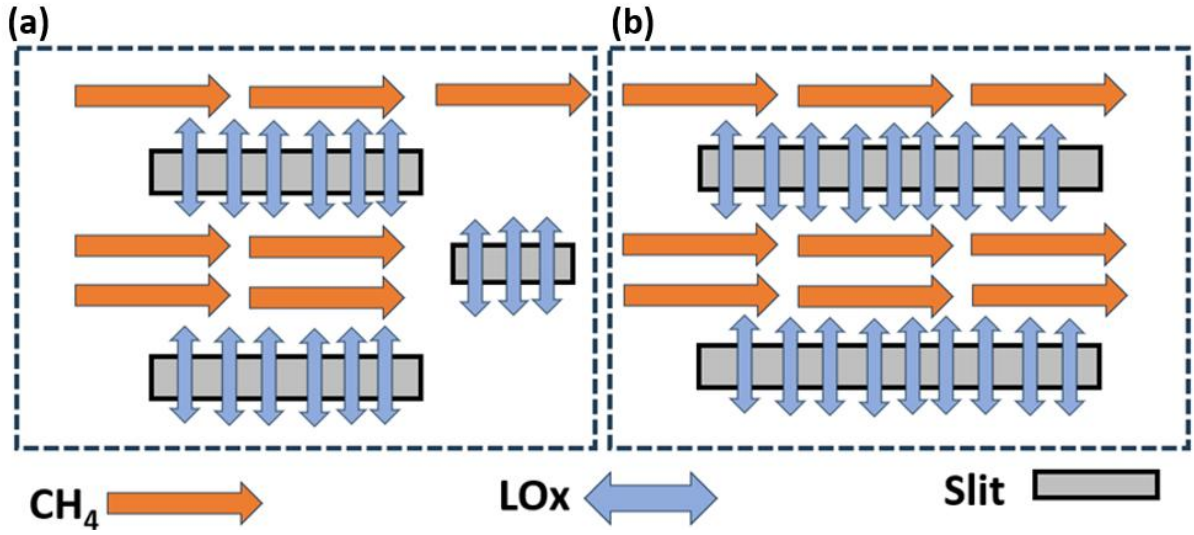

Fig.32. Schematic of mixing mechanism a) double row slit, b) single row slit

At higher slit numbers, the overall slit width increases, reducing methane bypass. Consequently, the second row becomes less active, making a single row configuration more effective. Eliminating the second row and further increasing the slit height enhances mixing performance. Slits with a higher height to width ratio (h/w) improve the interaction between LOx and methane along the combustor. For N = 22, the single row configuration has a higher h/w of 10.1 compared to 6.16 for the double row configuration, resulting in superior mixing. Reducing the number of slits concentrates the propellant streams, promoting stronger near injector interaction and faster temperature rise. For lower slit numbers, the overall slit width decreases, allowing more methane to bypass the slits and flow toward the pintle, where additional mixing occurs and pintle temperature increases. In this case, the double row configuration is advantageous, as the bypassed methane mixes with the LOx from the second row, reducing methane flow toward the pintle and lowering the risk of excessive pintle temperature.

6.5.2 Pintle tip heating

Pintle tip heating is a commonly observed issue in pintle injectors, primarily caused by the high-temperature zone near the tip due to local combustion [29]. This phenomenon arises due to flame anchoring and the formation of recirculation zones near the pintle tip[18, 30]. From the previous section 6.5 , at lower number of slits (N=16, 18), it is observed that the mixing performance between single row and double row slits is similar. But it is observed that the pintle tip temperature for single row slits is higher compared to the double row of slits. As the number of slits in each row decreases, the temperature near the pintle is increasing. Fig.32shows the schematic of methane flow in the double row of slits and single row of slits. The reason for the high temperature near the pintle is the methane bypassing through the gaps between the slits in the single row is getting combusted near the pintle tip resulting in high temperature at the pintle. Whereas, in double row of slits the smaller slits are placed on bottom row and are placed in between the gaps of the higher slits in the top row, this helps in restricting the methane flow towards pintle tip as the escaping methane is further exposed to oxygen coming from the second row of slits further diverting the methane flow away from the pintle tip[31]. High temperature near the pintle tip causes thermal damage and reduces the

pintle life time. At higher number of slits, the blockage factor is high even in single row of slits, so the high temperature zone near the pintle is not observed as the lower slits are less active at N= 22,20 compared to N=18,16.

Fig. ***33***(a) shows the temperature plot along the central line for single row of slit configuration for different number of slits. The same trend is followed in single row of slits configuration as of double row of slit configuration; the decrease in number of slits increased the temperature near the pintle tip. It is observed that for N=22 configurations the pintle tip temperature is observed to be less than 400K, for both single and double row of slits. Fig. ***33***(b) represents the plot comparing the double row of slit to single row of slits for N=20 and N=16 configuration. It is observed that in both the cases, single slit have a slight higher temperature near the pintle tip.

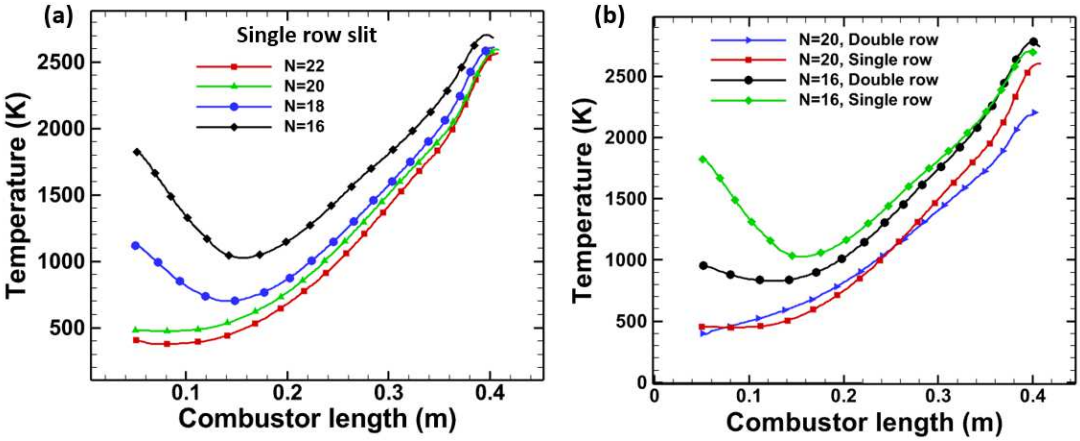

Fig. 33 Temperature along central line (a) comparing different slit configuration for single row of slits (b) comparison of single and double row slits for N=20,16

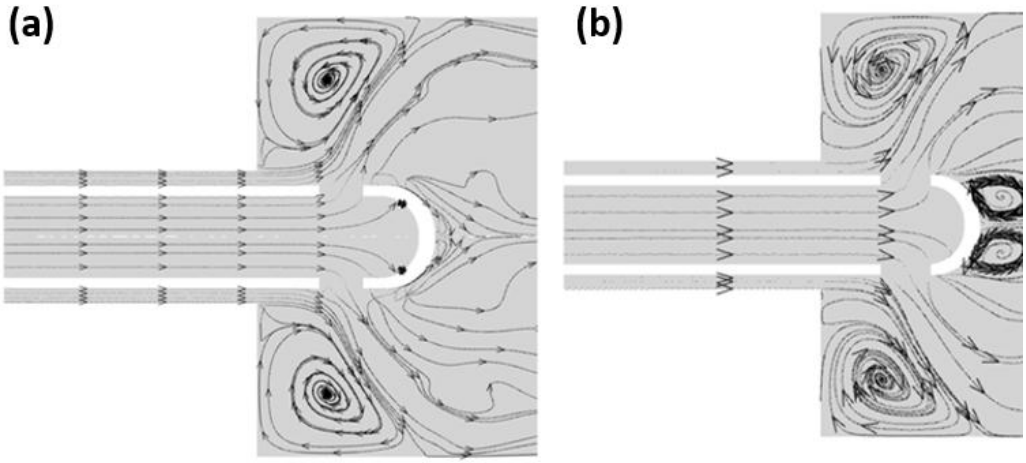

Fig.34 Stream line of N=20 fora) double row, b) single row of slits

Fig.34 presents the streamlines for single and double row slit configurations at N = 20. In the double row slits, recirculation zones are observed only near the walls at the combustor entry region, whereas in the single row slits, an additional recirculation zone forms at the pintle tip. This indicates methane bypassing, which flows toward the pintle tip and creates recirculation, where mixing occurs and further increases the pintle tip temperature [32, 33]. Fig.35 shows the pintle injector wall and tip temperature contours it is observed that lower slit count the pintle wall and tip temperature is high for both double row and single row

configurations. Compared to double row of slits, single slit has lesser pintle wall and tip temperature.

Overall higher slit height is preferred for better mixing. However, as the number of slits decreases, the methane bypass increases, which leads to the formation of high temperature zones near the pintle in the single row configuration. In such cases, the double row slit arrangement becomes preferable, as the second row promotes secondary mixing and limits methane flow toward the pintle tip. Therefore, it can be concluded that for lower slit numbers, a double row configuration is more effective, whereas for higher slit numbers, a single row configuration with increased slit height is more suitable, as it provides a larger jet interaction area and improved mixing performance.

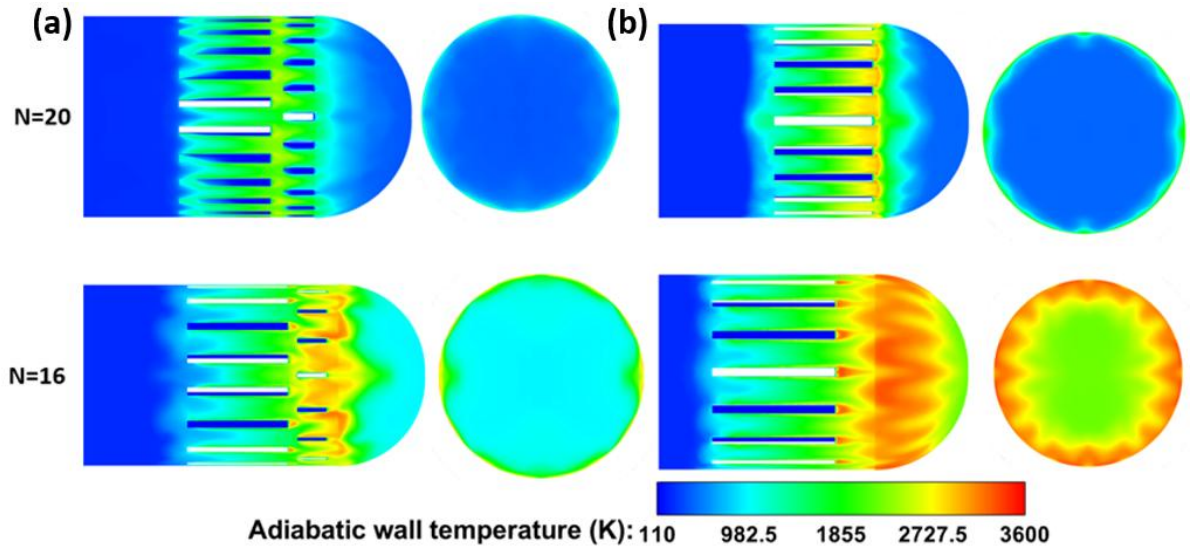

Fig.35 Pintle injector wall and tip (radialview) temperature contour for a) double row, b) single row of slits

7 Conclusion

An analytical model is developed for designing a pintle injector for a high-pressure LOx-methane engine, and a CFD based design methodology is employed to optimize the injector prior to experimental cold flow and hot fire testing. The initial injector configuration exhibits poor mixing, characterized by weak flame development, an extended low temperature region in the combustor, and a high oxygen mass fraction at the throat, indicating incomplete combustion. Parametric analysis shows that variations in annular gap and skip ratio do not result in significant improvements in mixing behaviour.

The present study shows the influence of slit geometry on the mixing and combustion performance of the pintle injector. For double row slit configurations, a reduction in the number of slits leads to a decrease in the length of the oxygen rich core, indicating stronger interaction between the central LOx jet and the surrounding methane annulus. Although the 16slit configuration exhibits high mixing intensity, it also results in the formation of a high temperature zone near the pintle tip, which poses thermal management challenges. Whereas the 18,20slit double row configuration shows better mixing while maintaining a low temperature region near the pintle, eliminating the need for additional cooling and improving pintle durability and resistance to thermal stresses.

A comparison between single row and double row slit configurations further reveals that the jet interaction area is a key parameter governing mixing. For higher numbers of slits (N = 22), the single row configuration provides improved mixing compared to the corresponding double row case due to increased slit height, which enhances the interaction between the LOx jets and the methane annular flow. Additionally, the higher number of slits at N=22 reduces methane bypassing, thereby avoiding the formation of high temperature zones near the pintle. However, as the number of slits decreases, methane bypassing in the single row configuration increases leading to high temperature region near the pintle. Analysis shows that a single row slit configuration can be effectively employed without the complexity of a two row slit configuration. CFD analysis reveals the efficacy of the single row slit design and demonstrates that CFD based design optimization can be utilized for complex injector designs in high performance LOx-methane engines.

Overall, the results demonstrate that mixing efficiency is predominantly controlled by slit geometry, with minimal influence from annular gap and skip ratio within the tested range. For higher slit numbers, a single row configuration with increased slit height is preferable due to enhanced jet interaction, whereas for lower slit numbers, a double row configuration can offer a better balance between mixing performance and thermal management. In future work, cold flow tests will be conducted to further study the effects of slit geometry, skip ratio, and annular gap, providing a clearer and more detailed understanding of their individual and combined influence on flow behaviour and mixing characteristics.

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